



Expansion tank with bursting diaphragm 600-0002, 600-00027

Operation Manual

Read this manual completely and keep it!

No warranty in case of damages caused by incorrect operation.

Imprint

**Expansion tank
with bursting diaphragm
600-0002, 600-00027**

Operation Manual

Version 1.0

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Subject to change.

Installation Data



NOTE

Please fill in this form following commissioning. It will help you and your GRUNDFOS Alldos servicing partner to adjust the device during subsequent corrections.

Owner:

GRUNDFOS Alldos customer No.:

Contract No.:

Order No. of device:

Serial number of device:

Put into service on:

Location of device:

Used for:

Installation Diagram

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1 Of General Interest

1.1 Structure of the Documentation

The documentation of GRUNDFOS Alldos Expansion tank with bursting diaphragm, 600-0002, 600-00027 consists of the:

- ❑ the *Operation Manual*:
 - technical data
 - instructions for commissioning and operation
 - specific safety instructions
 - instructions for maintenance and repair

If you require further information or if any problems arise which are not reflected in detail in this manual, contact GRUNDFOS Alldos directly for the information needed.

1.2 Using this Paper

- ❑ Descriptions are written as plain text.
- ❑ Lists are indicated by squares (❑), sublists by dashes (-).
- ❑ Operation steps are indicated by bullets (●), substeps by small bullets (•).
- ❑ Cross-references are indicated by *italic letters* and an arrow (→).
- ❑ The headings **WARNING**, **CAUTION** and **NOTE** have the following meaning:



WARNING ***Danger of injuries and accidents!***



CAUTION ***Danger of incorrect operation or damage to the product!***



NOTE ***There is an exceptional feature.***

1.3 Warranty

The warranty period is 24 months from the date of delivery by GRUNDFOS Alldos.

Warranty in accordance with our general terms of sale and delivery shall only be valid, if

- ❑ the product is used according to the information within this paper,
- ❑ the product is not being opened or used improperly,
- ❑ maintenance and repair is implemented exclusively by authorized and qualified personnel,
- ❑ original spare parts are used for repairs,
- ❑ only parts approved by GRUNDFOS Alldos are being used in the whole gas dosing plant.

Warranty is not valid for typical wearing parts, e. g.

- ❑ gaskets, bursting disk

2 Safety Instructions



WARNING

Before commissioning, read the GRUNDFOS Alldos information paper "Function and Safety of Chlorination Plants".

2.1 Application of the Product

The GRUNDFOS Alldos expansion tank with bursting diaphragm is a safety device for the case of too high pressure in the lines.

The permissible media are indicated on the type plate.



WARNING

Other applications are not intended and are not permitted. ALLDOS Eichler GmbH accept no responsibility for damages caused by unintended use.

2.2 Obligations of the Operating Authority

The operating authority of the plant is responsible for

- ☐ obeying the following regulations (in Germany):
 - Unfallverhütungsvorschrift *Chlorung von Wasser* (GUV-V D5)
 - Unfallverhütungsvorschrift *Druckbehälter* (GUV 2.6)
 - Unfallverhütungsvorschrift *Gase* (GUV 9.9)
 - Verordnung über Arbeitsstätten (ArbStättV)
 - Verordnung über gefährliche Arbeitsstoffe (ArbStoffV)
 - obeying local safety regulations, if necessary
- ☐ instructing the operation personnel
- ☐ providing the stipulated safety equipment (→ *information paper "Function and Safety of Chlorination Plants"*)
- ☐ arranging regular maintenance

2.3 Averting Dangers



WARNING

Cleaning, maintenance and repair only by authorized personnel!

Danger of gas breakout!

3 Technical Data

3.1 General Data

	<i>Order no.</i>	<i>medium</i>	<i>volume</i>
	96706391 (600-0002)	Cl ₂	10 litres
	96729960 (600-00027)	NH ₃	10 litres
Inlet connection	G1" (female thread)		
Max. operation pressure	12.6 bar		
Testing pressure	23 bar		
Bursting pressure	16 - 17.6 bar		
Permissible ambient temperature	0 to +40 °C (Cl ₂ , NH ₃)		

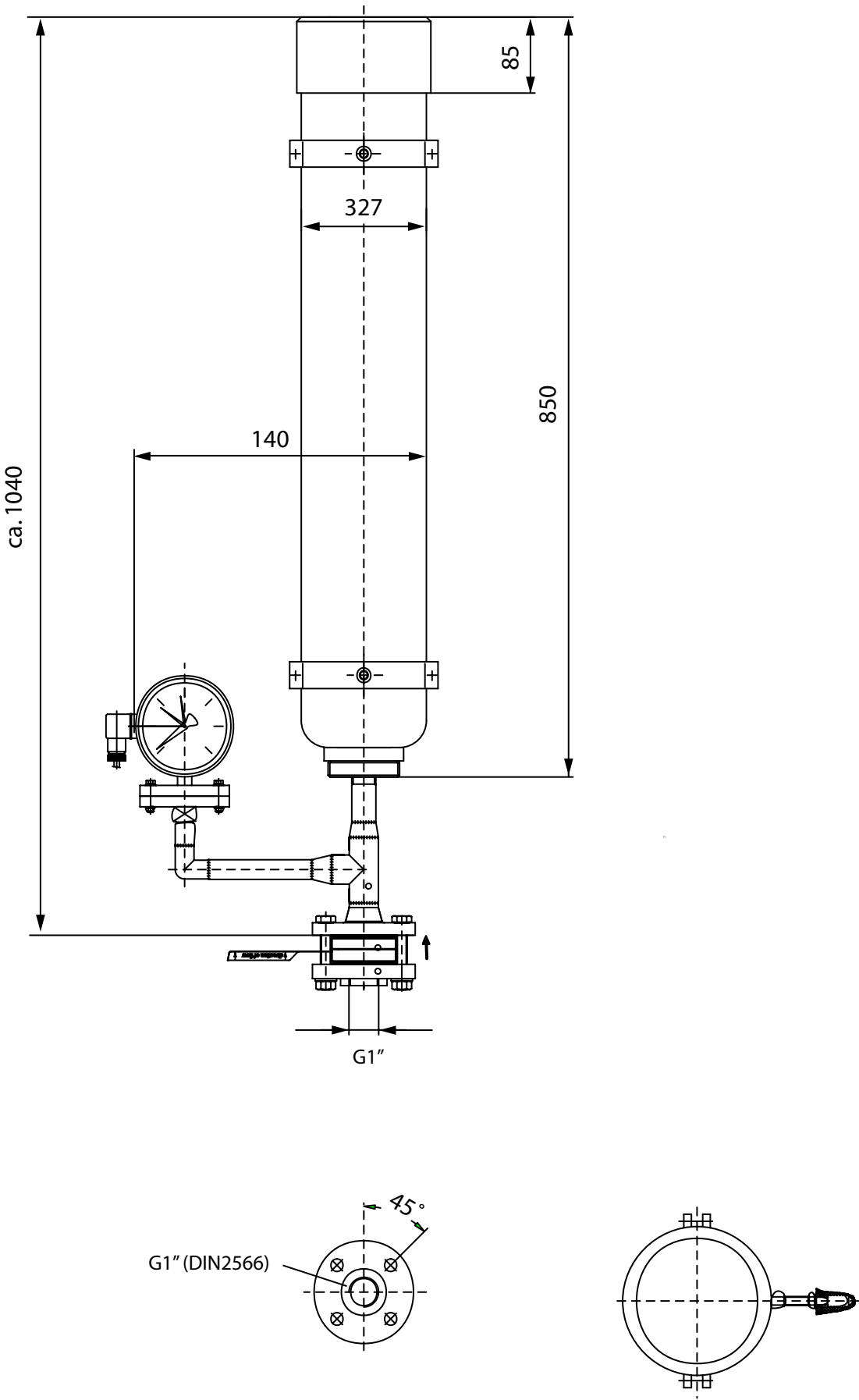
3.2 Materials

Expansion tank	Tempering steel
Further parts in contact with media	Steel, stainless steel, Cupric nickel alloy

3.3 Electrical Data

Contact load of the manometer	max. 250 V, max. 50 VA ohmic load
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3.4 Dimensioned Drawing



4 Installation

4.1 Transport and Storage

- Handle with care, do not throw!
- Dry and cool storage place.
- The gas connection must be tightly closed during transport and storage!



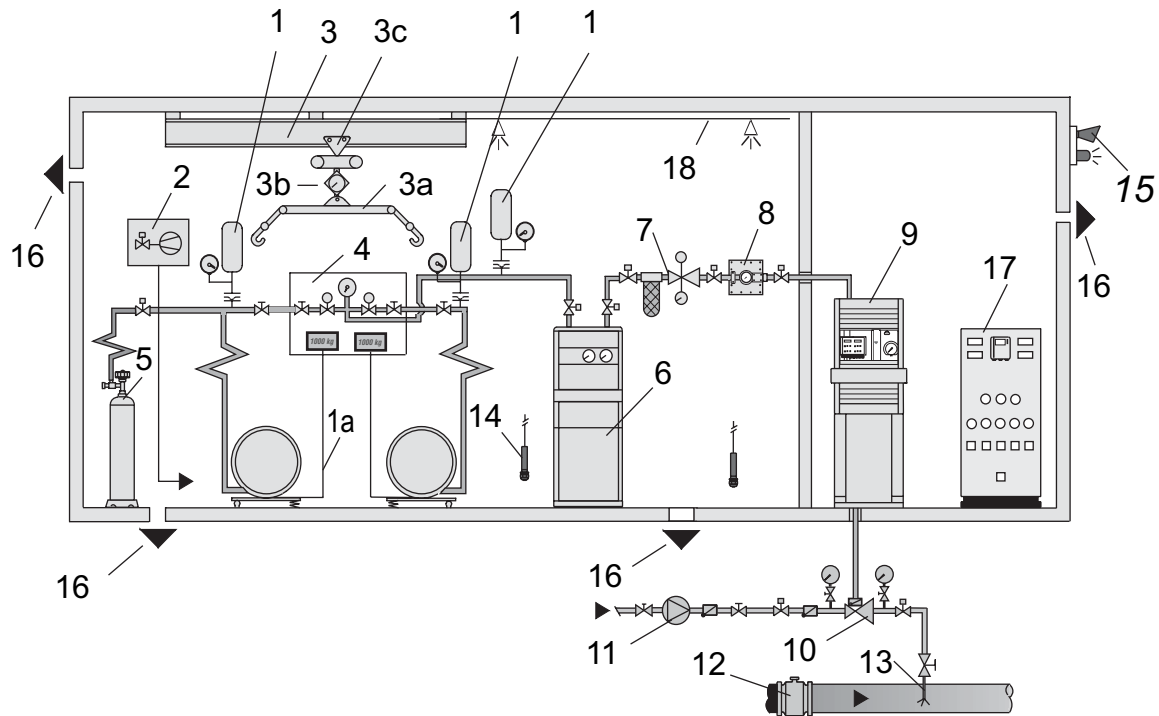
WARNING

***If humidity penetrates the device:
Danger of gas breakout!***

4.2 Unpacking

- Observe when unpacking:
 - No humidity should get into gas-leading parts!
 - No foreign matter should get into gas-leading parts!
- Mount as soon as possible after unpacking.

4.3 Installation Example



- 1 Expansion tank
- 1a Chlorine drum balance
- 2 Control for the pneumatic chlorine drum quick-action stop valves
- 3 Chlorine drum lifting device
- 3a Lifting beam
- 3b Spring balance for cranes
- 3c Chain hoist
- 4 Automatic change-over device
- 5 Nitrogen rinsing device
- 6 Chlorine evaporator
- 7 Pressure reducing valve and Filter
- 8 VGS 146-148 vacuum regulator
- 9 VGS 141-145 gas dosing system
- 10 High-capacity injector
- 11 Motive water booster pump
- 12 Flowmeter with 4-20 mA output
- 13 Injection unit
- 14 Chlorine gas sensor (ALLDOS safety system Gasdetect)
- 15 Optical and acoustic alarm devices
- 16 Suction nozzle to the chlorine gas neutralization system
- 17 Central control panel with computer interface
- 18 Sprinkling system

4.4 Mounting



WARNING

Observe information paper "Function and Safety of Chlorination Plants"!



CAUTION

Only screw down the device to perfect flat and smooth surfaces.

- Mark the bore holes of the clamp (supplied).
- Bore the holes.
- Put in dowels.
- Hang up the expansion tank at the desired place.

5 Commissioning

5.1 Preparation for Commissioning

5.1.1 Gas Connections



WARNING

Before connection: Ensure that the valves of all gas containers are closed.

Only use the intended line types!

All lines must be stress-free!

- Connect the supply line (coming from the gas container or header line) to the inlet flange.
Permissible line type:
 - seamless drawn steel pipe according to DIN 2441, G 1"
 - free of oil, grease and rust deposits

5.1.2 Electrical Connections



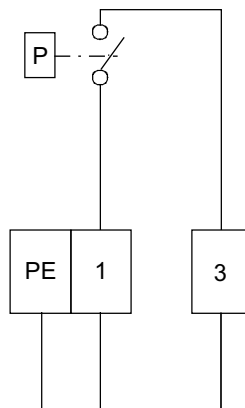
WARNING

Electrical connections may only be established by authorized, qualified personnel!

Observe local safety regulations when selecting the type of cables!

- Connect the contact lines of the manometer to the control panel following the connection chart.

Berstscheibe gebrochen
Bursting disk broken
Rupture du disque d'eclatement



Verbindung zum Schaltschrank
Connection to the control panel
vers armoire de commande

5.2 Inspections before Commissioning

5.2.1 Checking the Settings

- Check the setting of the manometer, correct it, if necessary:
 - Contact manometer behind the bursting disk: 1 bar

5.2.2 Checking the Electrical Connections

- Check if all electrical connections are correct.

5.2.3 Checking the Tightness

Check tightness of the **total plant** before start-up.



WARNING

Check the tightness not until the total plant is ready for start-up.

Danger of gas breakout!

Put on safety equipment (gas mask) during the tightness check!

- If the plant is equipped with a nitrogen rinsing device: Check tightness roughly with nitrogen (for all dosing media).
- Detailed checking:
 - with burning sulphur (dosing medium NH_3)
 - with ammonia (dosing media Cl_2)

All dosing media: Checking the tightness with nitrogen

- Close all container valves
- Open container connection valves and all shut-off valves up to the gas dosing system
- Open the connection valve of the nitrogen cylinder
- Slowly open the valve of the nitrogen cylinder, until the lines have a pressure of about 10 bar (read at the manometer of the vacuum regulator).

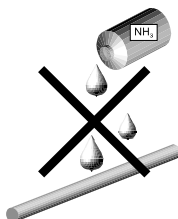


WARNING

Maximum nitrogen pressure 12.5 bar!

Danger of damages and gas leakage when being exceeded!

- Apply soap water to all components under pressure
 - Formation of bubbles and/or pressure drop at manometer: **Leakage in pressure lines!**
 - **Depressurize the plant!**
 - **Eliminate leakage!**
 - **Check tightness again!**
 - No formation of bubbles, pressure at manometer does not drop significantly within one hour: **Pressure lines are tight.**



WARNING

Liquid ammonia must not come in contact with parts of the plant!

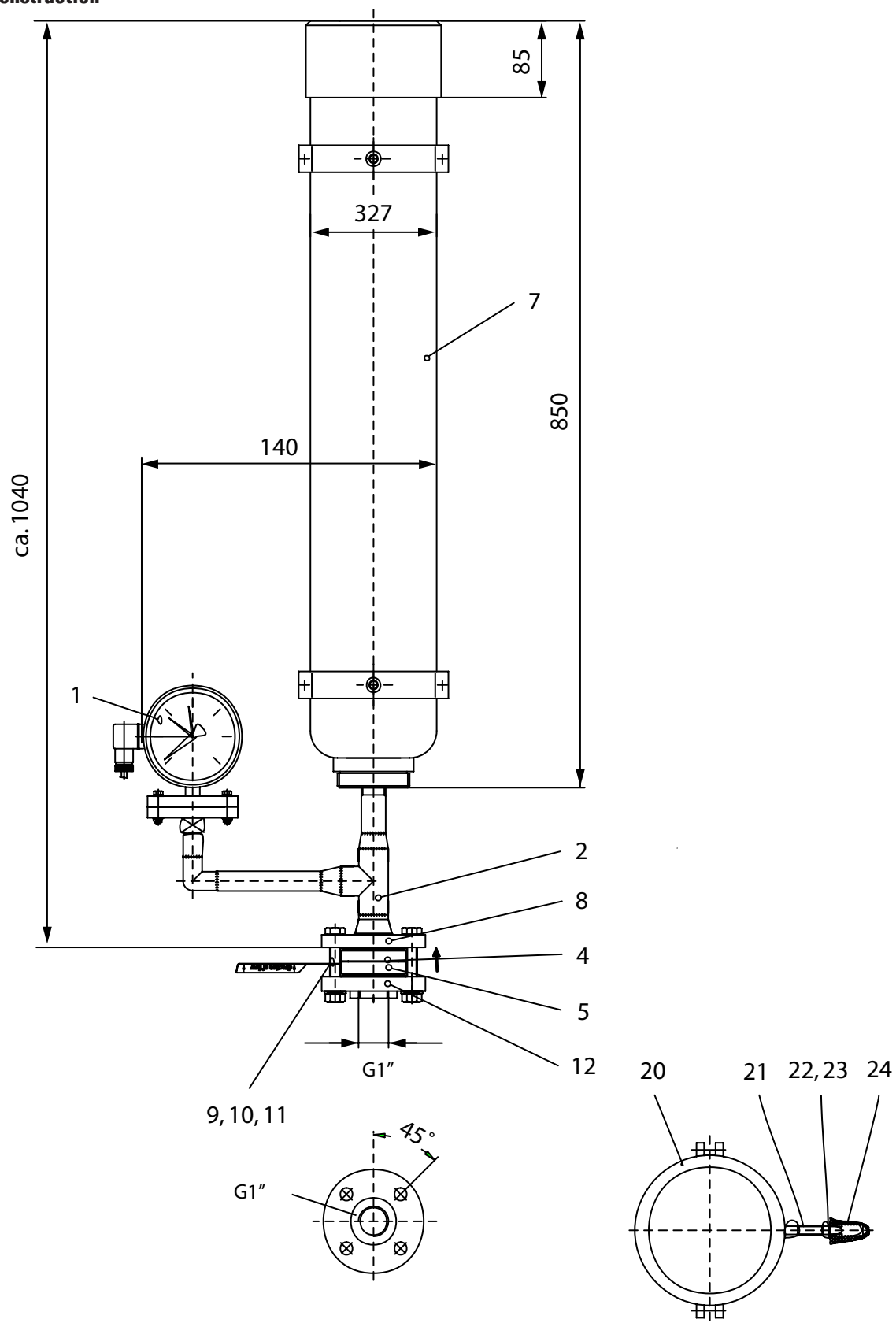
Danger of leakages by corrosion!

6 Operation

The Expansion tank with bursting diaphragm requires no operating.

6.1 Description of the Device

6.1.1 Construction



- 1 Contact manometer
- 2 Pipework
- 4 Bursting disk
- 5 Support for bursting disk
- 7 Expansion tank
- 8 Flat gasket
- 9 Washer
- 10 Nut
- 11 Screw
- 12 Threaded flange
- 20 Clamp
- 21 Threaded pin
- 22 Nut
- 23 Washer
- 24 Anchor

6.1.2 Function

The expansion tank is a safety device for too high pressures. It prevents the destruction of liquid gas lines (Cl_2 and NH_3) by too high pressures. Closed valves together with a high room temperature may be the cause of the too high pressure. Closed valves are a fault in operation; they prevent pressure compensation.

6.2 Possible Failures

6.2.1 Damaged Bursting Disk

If the bursting disk is damaged, this will be displayed by the corresponding contact manometer.

- Switch off the whole plant
- Replace the bursting disk

WARNING

***Under no circumstances the plant may be operated with a damaged bursting disk!
Danger of gas breakout!***



7 Maintenance

We recommend to exchange the bursting disk every 24 months:



WARNING

Maintenance and repair only by authorized personnel!

***Switch off the whole plant before doing maintenance work!
Danger of gas breakout!
Switch off mains voltage!***

***Check the tightness before restarting!
Danger of gas breakout!***

7.1 Replacing the Bursting Disk

If the bursting disk is damaged, it must be replaced immediately.



WARNING

***Under no circumstances the plant may be operated with a damaged bursting disk!
Danger of gas breakout!***

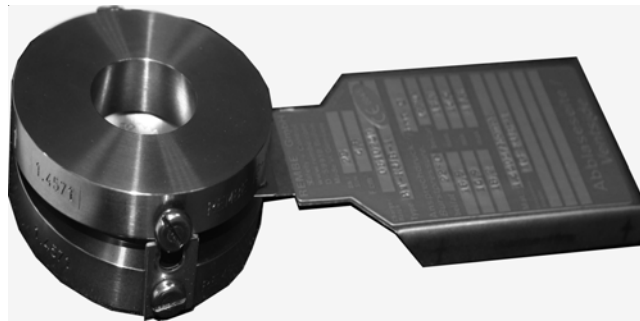
Required tools and accessories:

- ☐ bursting disk 95705473 (12.6928-1)

- ❑ 2 flat gaskets 96728864 (54.030)
- ❑ 2 fork wrenches 19 mm
- ❑ medium-size screwdriver

While the plant is still running:

- Close all container valves
- Let the system run until the manometer at the expansion tank shows 0 bar.
If the system is equipped with a nitrogen rinsing device:
 - Rinse with nitrogen for about 5 minutes (the nitrogen will be sucked into the water to be treated).



- Loosen the screws of the flange connection.
- Take out the support of the bursting disk and unscrew it.
- Put in a new bursting disk and screw it together.



WARNING *Observe mounting position!*

- Replace the flat gaskets on both sides of the support.
- Place the support and tighten the flange connection.
- Restart the Expansion tank as described for commissioning.

8 Spare Parts

We suggest to store a bursting disk and flat gaskets.

Order no.	Description
95705473 (12.6928-1)	Bursting disk
96728864 (54.030)	Flat gasket

